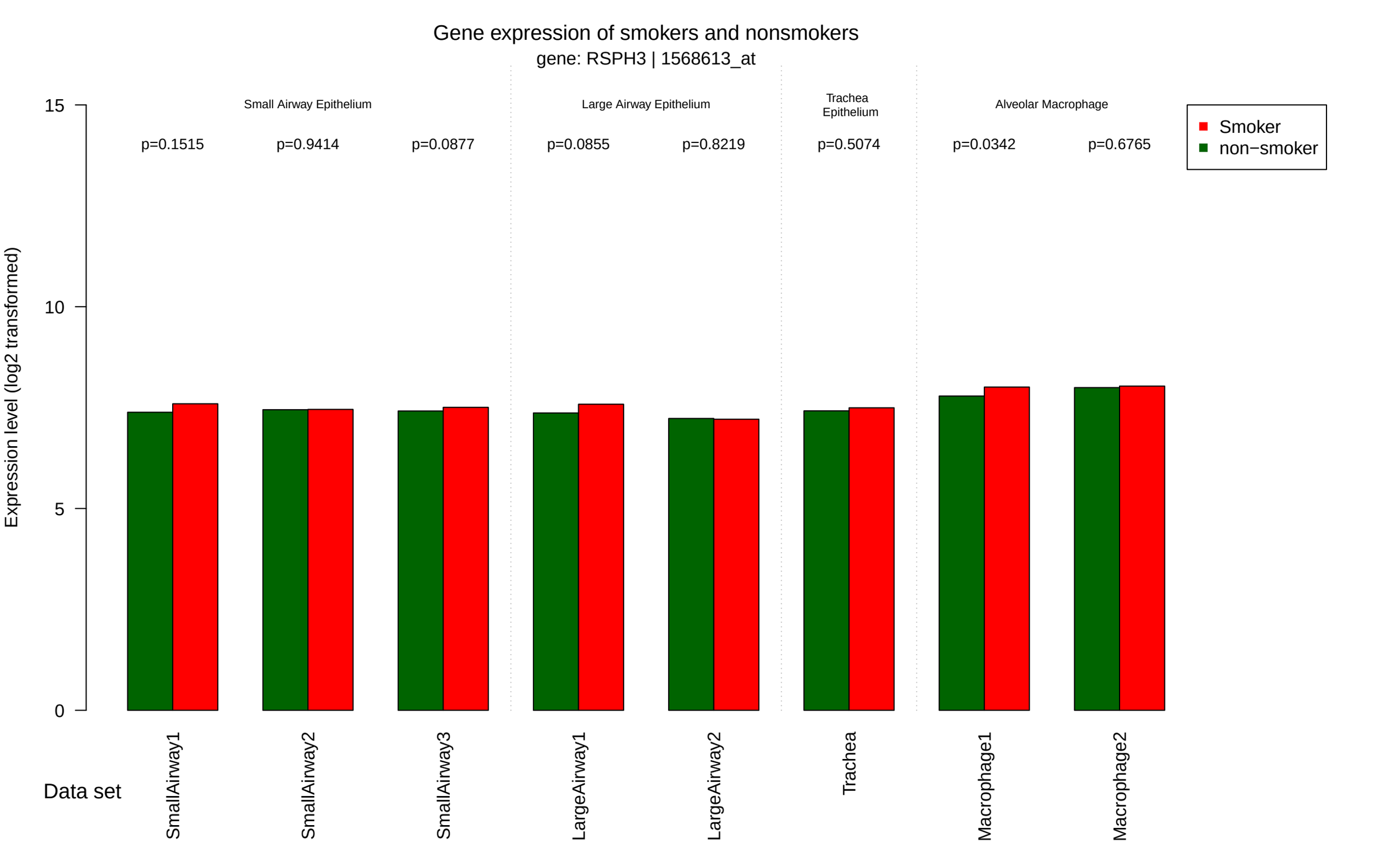
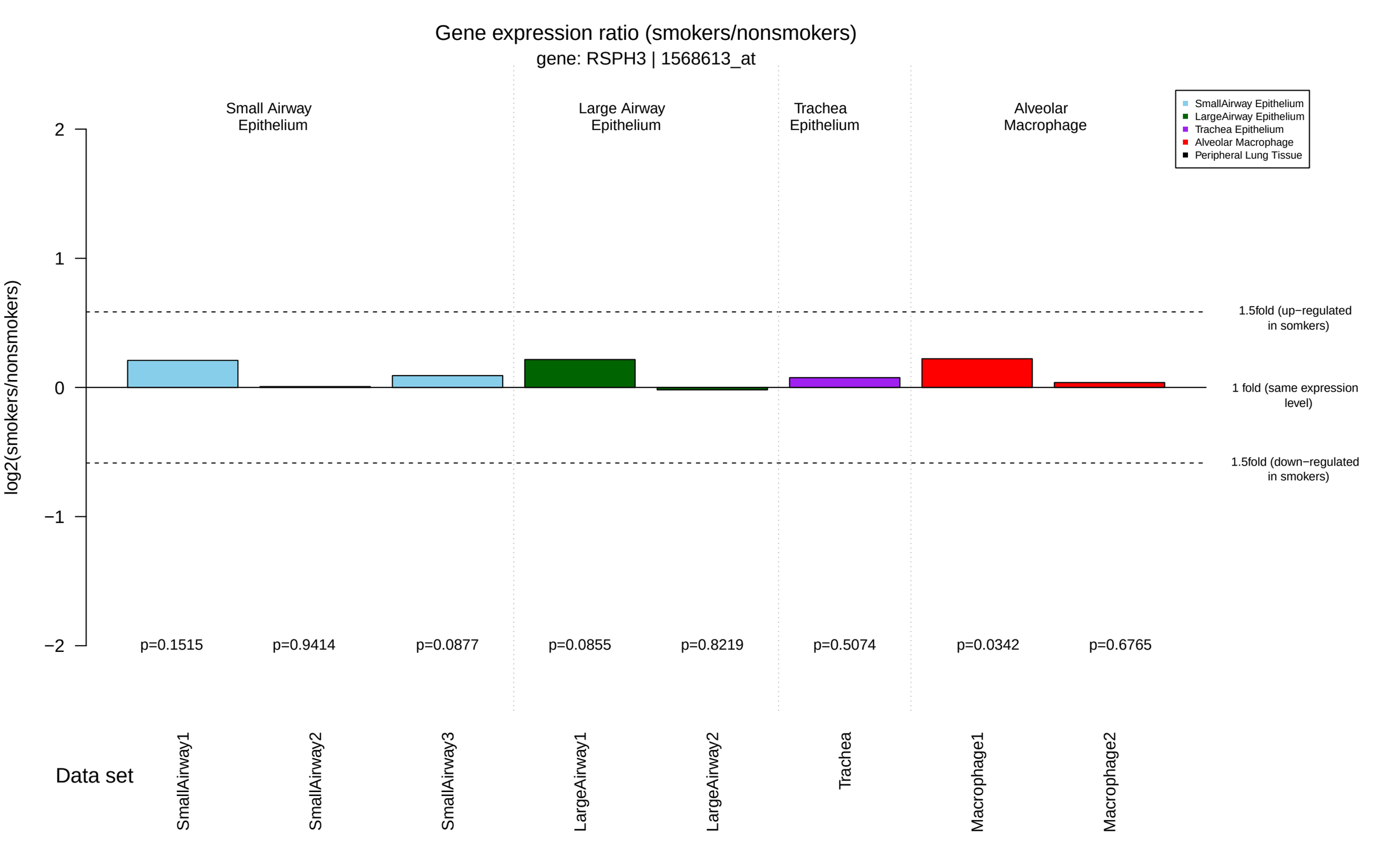
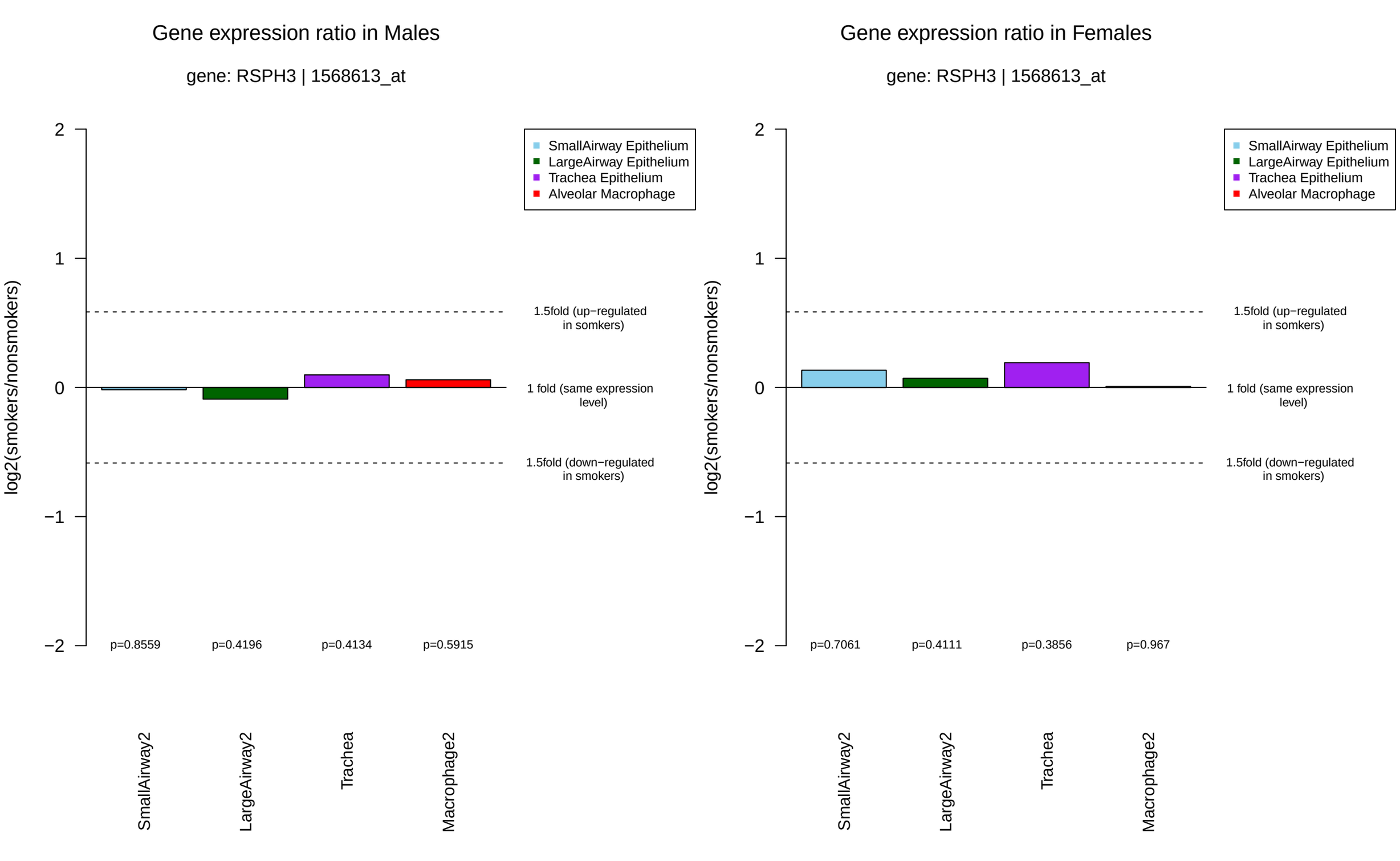
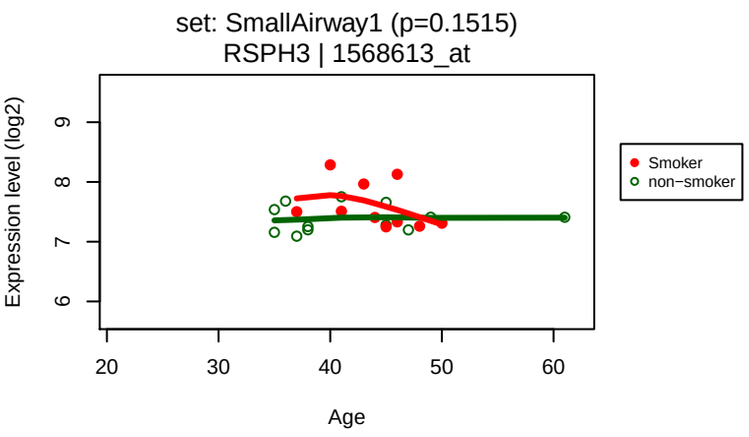
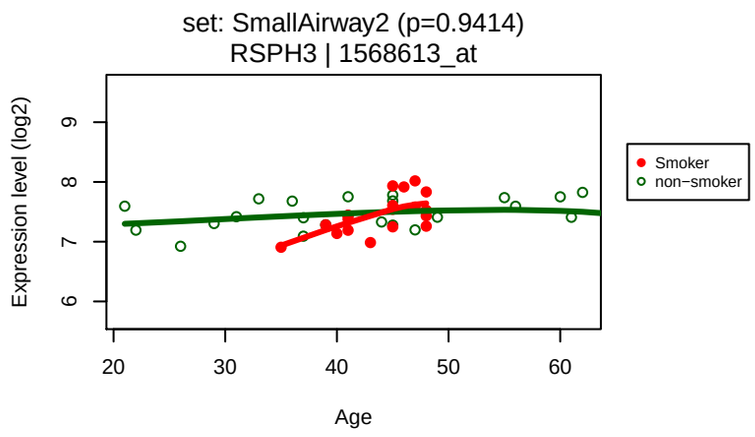
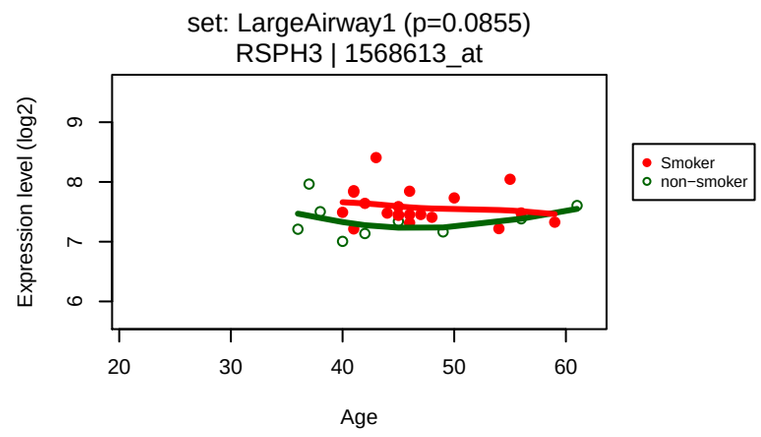

Gene expression of smokers and nonsmokers
gene: RSPH3 | 1568613_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1515
p=0.9414
p=0.0877
p=0.0855
p=0.8219
p=0.5074
p=0.0342
p=0.6765
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

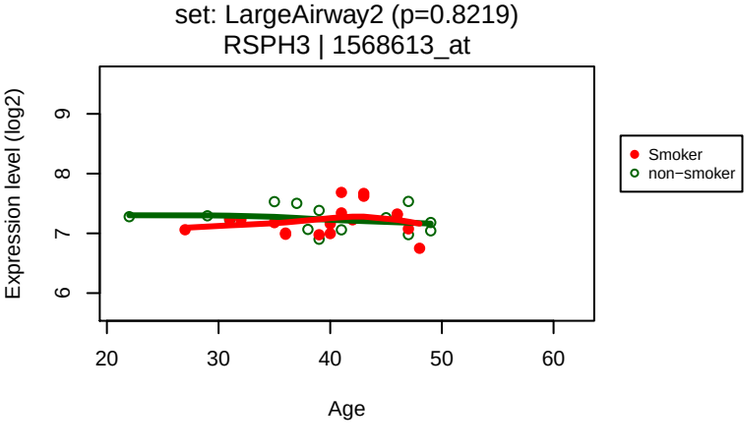

Gene expression ratio (smokers/nonsmokers)
gene: RSPH3 | 1568613_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1515
p=0.9414
p=0.0877
p=0.0855
p=0.8219
p=0.5074
p=0.0342
p=0.6765
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

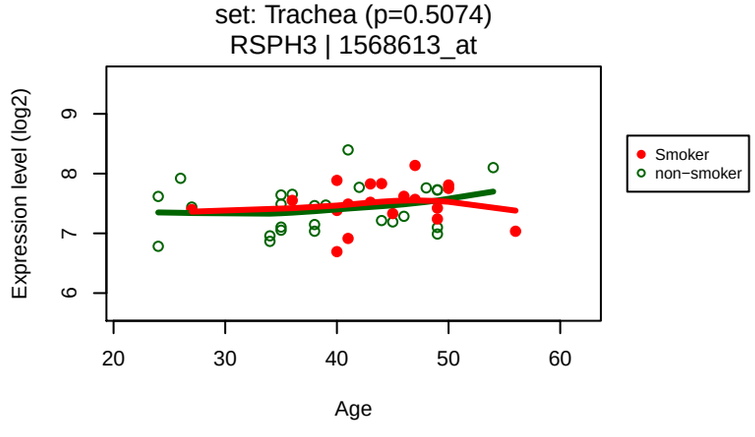

Gene expression ratio in Males
gene: RSPH3 | 1568613_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8559
p=0.4196
p=0.4134
p=0.5915
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSPH3 | 1568613_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7061
p=0.4111
p=0.3856
p=0.967
SmallAirway2
LargeAirway2
Trachea
Macrophage2

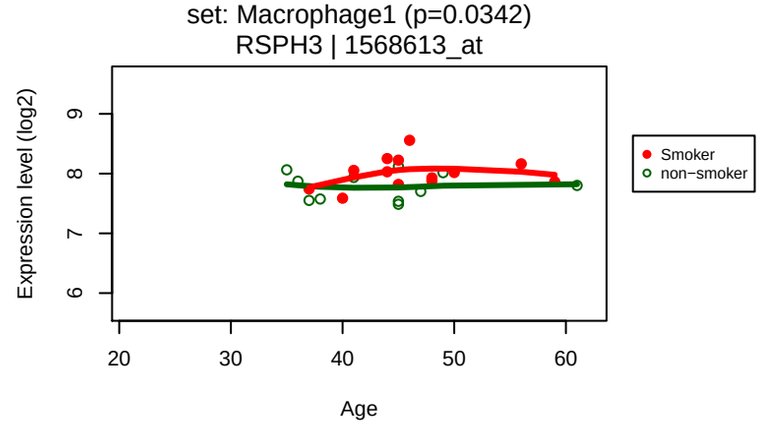

set: SmallAirway1 (p=0.1515)
RSPH3 | 1568613_at
Expression level (log2)
Age
Smoker
non−smoker

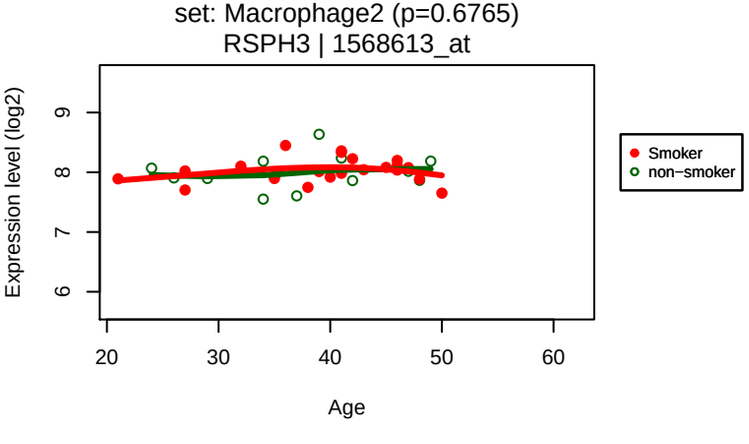

set: SmallAirway2 (p=0.9414)
RSPH3 | 1568613_at
Expression level (log2)
Age
Smoker
non−smoker

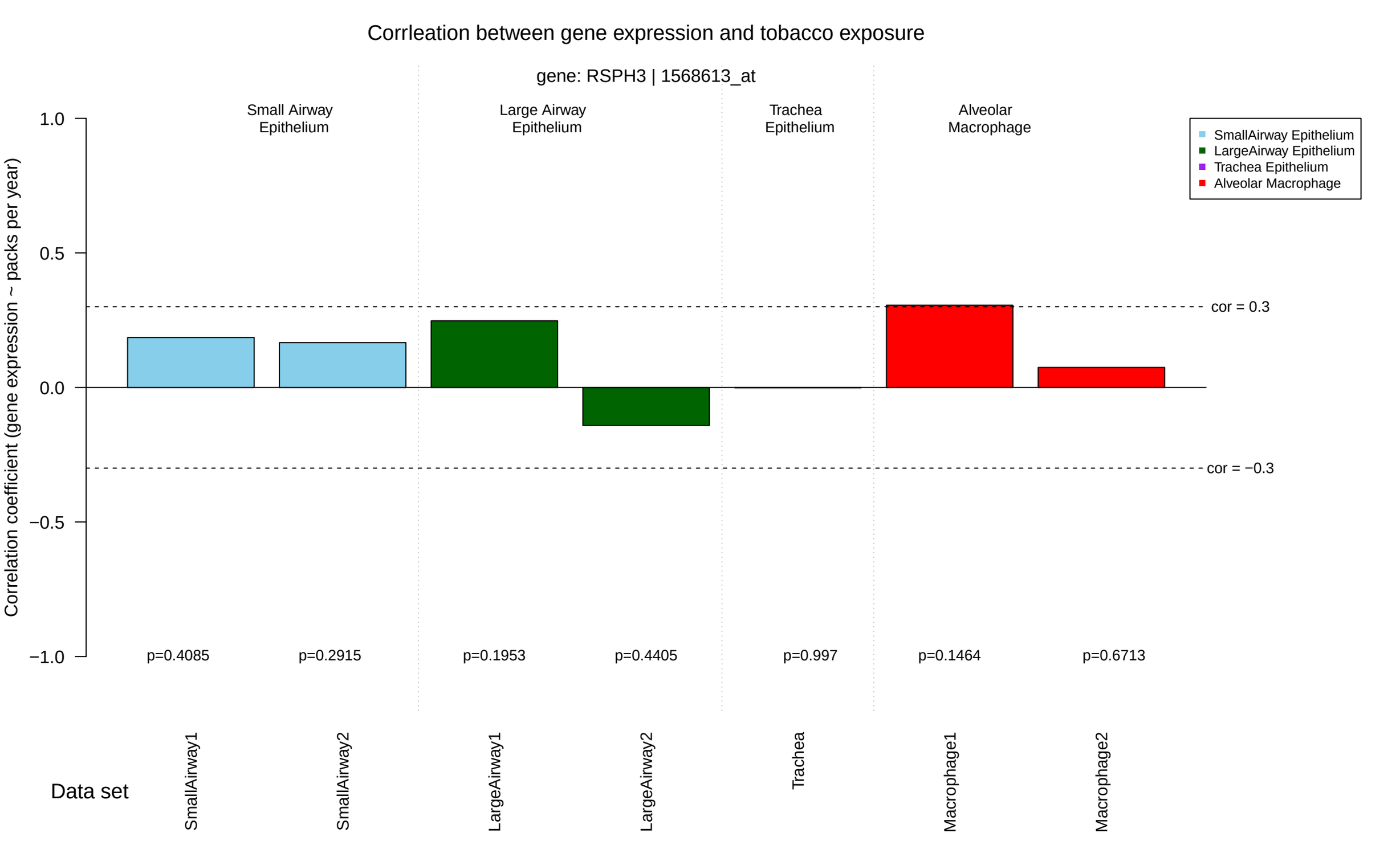

set: LargeAirway1 (p=0.0855)
RSPH3 | 1568613_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8219)
RSPH3 | 1568613_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5074)
RSPH3 | 1568613_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0342)
RSPH3 | 1568613_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6765)
RSPH3 | 1568613_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSPH3 | 1568613_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4085
p=0.2915
p=0.1953
p=0.4405
p=0.997
p=0.1464
p=0.6713
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2